

1. Which algorithmic technique solves a problem by breaking it into smaller subproblems and combining the results?

- A. Greedy Algorithm
- B. Dynamic Programming
- C. Brute Force
- D. Backtracking

Answer: B

2. What will be the output of the following code?

```
int sum = 0;
for(int i = 1; i <= 5; i++)
    sum += i;
printf("%d", sum);
```

- A. 10
- B. 15
- C. 20
- D. 5

Answer: B

3. Which data structure follows the Last In, First Out (LIFO) principle?

- A. Queue
- B. Linked List
- C. Stack
- D. Tree

Answer: C

4. What is the time complexity of a binary search algorithm?

- A. $O(n)$
- B. $O(\log n)$
- C. $O(n^2)$
- D. $O(1)$

Answer: B

5. What will be the output?

```
x = [1, 2, 3]
print(x[-1])
```

- A. 1
- B. 2
- C. 3
- D. Error

Answer: C

6. Which loop is guaranteed to execute at least once?

- A. for loop
- B. while loop
- C. do-while loop
- D. Nested loop

Answer: C

7. Which of the following is an example of recursion?

- A. Using a for loop to calculate a factorial
- B. A function calling itself to calculate a factorial
- C. Using an array to store values
- D. Sorting with bubble sort

Answer: B

8. What is the output of the following code?

```
int a = 10;
if(a > 5)
    printf("A");
else
    printf("B");
```

- A. A
- B. B
- C. AB
- D. Error

Answer: A

9. Which sorting algorithm has the best average-case time complexity among the following?

- A. Bubble Sort
- B. Selection Sort
- C. Merge Sort
- D. Insertion Sort

Answer: C

10. Which approach tries all possible solutions until the correct one is found?

- A. Divide and Conquer
- B. Dynamic Programming
- C. Brute Force
- D. Greedy Algorithm

Answer: C